

ViPragSent: XLM-R-large for Vietnamese Pragmatic Sentiment Analysis

Anonymous ACL submission

Abstract

Pragmatic sentiment in Vietnamese text is not exhausted by polarity. Sarcasm, irony, idiomaticity, implicitness, code-switching, and mocking can change how a reader should interpret an utterance even when its surface sentiment is positive or negative. We introduce ViPragSent, an XLM-R-large encoder with six distinct binary pragmatic heads, separate three-way polarity and seven-way emotion auxiliaries, and a training-only teacher-forced rationale objective. The model uses a shared first-nonpadding-token representation and homoscedastic uncertainty weighting to couple these tasks while keeping their semantic roles separate. On a matched 2,000-example ID/gold test cohort, ViPragSent reaches 93.67 ± 0.19 macro-pragmatic F1 versus 92.93 ± 0.14 for the standard XLM-R baseline, an observed +0.73 percentage-point difference. Its mean is also highest on all six pragmatic heads and on the macro average among five complete three-seed baseline families. Follow-up analyses show that the full objective is an operating point rather than a uniform winner: Q2 reports accuracy, calibration, and cost trade-offs; Q3 reports a non-monotonic positive-label budget curve with contextual baselines; and Q4 reports target ECE 0.0386 ± 0.0069 versus Vistral 0.0345 ± 0.0041 . These results define what the implemented model contributes while keeping rationale supervision separate from inference.

1 Introduction

Sentiment analysis is often framed as a polarity problem, but pragmatic meaning can reverse or qualify the interpretation suggested by a surface-positive or surface-negative expression. A Vietnamese utterance may be sarcastic, ironic, idiomatic, implicit, code-switched, or mocking without belonging to an emotion

category. These phenomena are therefore useful prediction targets in their own right. Treating them as six emotion classes would collapse distinct communicative functions and obscure the role of emotion as an auxiliary signal. ViPragSent adopts the narrower formulation: six pragmatic phenomena are six binary tasks, while polarity and emotion are separate auxiliary tasks.

Vietnamese offers a demanding setting. PhoBERT shows the value of language-specific pretraining (Nguyen and Nguyen, 2020), whereas XLM-R provides a multilingual transfer regime across one hundred languages (Conneau et al., 2020). ViSoBERT applies the XLM-R architecture to Vietnamese social-media tasks (Nguyen et al., 2023b). These contrasting settings motivate testing a large multilingual encoder on a bundle of pragmatic distinctions rather than one sentiment label.

The challenge is also objective design. Pragmatic labels may correlate with polarity and emotion without being identical. Multi-task learning can use related signals (Caruana, 1997), while uncertainty weighting adapts relative loss scales (Kendall et al., 2018). These are empirical choices, so we report the training losses, causal rationale decoder, and classification-head inference interface explicitly.

This paper studies ViPragSent XLM-R-large through four linked questions:

- **Q1a–Q1b:** How does the primary model behave in-domain, and how much of that behavior is retained on external Vietnamese sentiment and emotion datasets?
- **Q2:** What changes when the polarity, emotion, explanation, multitask, or task-uncertainty components are removed?

- **Q3:** How does pragmatic performance change as the number of positive examples is constrained while the evaluation procedure is held fixed?
- **Q4:** How well calibrated are the resulting confidence estimates relative to comparison models under a common expected-calibration-error protocol?

The contribution is a focused analysis of one primary XLM-R-large model rather than a leaderboard claim. We contribute (i) an implemented shared encoder with separated pragmatic, polarity, emotion, and teacher-forced rationale objectives; (ii) a six-head comparison on a matched ID/gold prediction cohort; and (iii) follow-up analyses that expose accuracy, calibration, cost, and budget trade-offs. All reported means summarize three saved runs, and comparison claims concern the evaluated prediction artifacts rather than a fresh training or inference rerun.

2 Related Work

2.1 Multilingual and Vietnamese encoders

XLM-R established a strong multilingual masked-language-model baseline by scaling pretraining across languages and evaluating cross-lingual transfer (Conneau et al., 2020). PhoBERT demonstrates the complementary case for a Vietnamese-specific encoder trained on word-segmented Vietnamese data (Nguyen and Nguyen, 2020). ViSoBERT moves toward Vietnamese social-media text while retaining the XLM-R architecture (Nguyen et al., 2023b). ViPragSent uses XLM-R-large as its primary encoder so that the study can focus on the pragmatic multi-task formulation and its auxiliary objectives.

The Q1a comparison contains standard XLM-R, PhoBERT, Sailor, and Vistral prediction artifacts. ViSoBERT is discussed here as contextual prior work, not as an evaluated Q1a baseline: no ViSoBERT prediction family is included in the verified comparison table. This distinction prevents a related model family from being mistaken for a measured result.

2.2 Pragmatic and affective supervision

The target formulation combines signals that are related but not interchangeable. UIT-VSFC provides Vietnamese student feedback with sentiment-oriented labels (Nguyen et al., 2018), whereas UIT-VSMEC is a Vietnamese social-media emotion corpus (Ho et al., 2020). ViPragSent uses these label families as auxiliary signals while retaining separate pragmatic outputs.

Sarcasm detection studies show why context can matter beyond the current utterance: conversation-aware models can outperform models that read only the current turn (Ghosh et al., 2018). A complementary line of work distills teacher-generated rationales as additional supervision in a multi-task objective (Hsieh et al., 2023); earlier rationale models likewise separate prediction from explanation (Lei et al., 2016). ViPragSent brings this rationale-supervision idea to a shared XLM-R-large encoder and tests it alongside Vietnamese pragmatic heads, while keeping rationale generation outside the deployed prediction interface.

2.3 Multi-task objectives, rationale supervision, and calibration

Multi-task learning shares a representation across related objectives and can transfer useful domain information between them (Caruana, 1997). The uncertainty-weighted objective follows the motivation that task losses need not contribute equally (Kendall et al., 2018). The implemented rationale decoder supplies a teacher-forced auxiliary loss during training and is discarded at inference.

Confidence is evaluated separately from discrimination: expected calibration error measures the gap between confidence and empirical correctness across bins, which can remain large when accuracy is strong (Guo et al., 2017). Q4 is therefore a protocol-specific calibration comparison, not a universal uncertainty claim.

3 Method

3.1 Task definition

Given a Vietnamese utterance x , ViPragSent predicts six binary pragmatic labels in the

manifest order: implicit sentiment, sarcasm, irony, idiom/figurative language, code-switching, and mocking. These labels represent distinct pragmatic phenomena. The model may also predict a three-way polarity label and a seven-way emotion label. The auxiliary labels are not additional pragmatic classes.

For each pragmatic head, the metric is binary macro-F1: the arithmetic mean of the F1 values for class 0 and class 1. The macro-pragmatic F1 is the arithmetic mean of the six binary macro-F1 values. This definition is important for the head-level table because it differs from a positive-class-only F1.

3.2 XLM-R-large encoder and prediction heads

The primary model uses FacebookAI/xlm-roberta-large at revision c23d21b0620b635a76227c604d44e43a9f0ee389. The encoder returns a hidden state for each input token. For the XLM-R family, the implementation pools the first nonpadding representation: the hidden state at the first position with an active attention mask, rather than a mean over the sequence. A 0.1 dropout layer feeds task-specific linear heads: six one-logit binary heads, one three-logit polarity head, and one seven-logit emotion head. At inference, predictions come from these classification heads.

3.3 Implemented auxiliary objective

The six pragmatic heads use weighted binary cross-entropy with logits, with positive-class weights computed from the training split. Polarity and emotion use class-weighted cross-entropy. For the full model, the eight classification losses have independent learned log-variance parameters. Let $\mathcal{T}$ be the eight classification tasks, let m_t be the recorded task multiplier, and let s_t be the learned log variance. The implementation clamps the log variance before applying uncertainty weighting:

$$\begin{aligned} s_t^c &= \text{clip}(s_t, -5, 5), \\ \mathcal{L}_{\text{cls+rat}} &= \sum_{t \in \mathcal{T}} \left[\frac{1}{2} e^{-s_t^c} (m_t \ell_t) + \frac{1}{2} s_t^c \right] \\ &\quad + \beta \mathcal{L}_{\text{rat}}, \quad \beta = 0.3. \end{aligned}$$

Here ℓ_t is the task loss before the multiplier, so the multiplier is applied inside the uncertainty-

weighted classification term. The rationale term is present only for the full rationale-enabled variant.

The rationale component is a two-layer causal **TransformerDecoder**. It uses a memory projection from the encoder hidden size to a decoder hidden size of 128, four attention heads, feed-forward size 512, dropout 0.1, causal masking, and tied token input/output embeddings. During training it consumes rationale tokens with teacher forcing and contributes token cross-entropy as $\beta \mathcal{L}_{\text{rat}}$ with $\beta = 0.3$. For the target run, the manifest identifies `approved_generated_rationales_train.jsonl` with 7,998 training rows. The inspected evidence does not identify a provider for these generated targets, so we treat this as a run-level training artifact rather than evidence about the original dataset’s annotation history. The decoder is discarded from the inference path and produces no reported prediction.

3.4 Training and selection

The frozen ViPragSent package is identified in the data handoff as the SEACrowd/ViSoBERT source package and contains 11,997 rows: 7,998 training, 1,999 development, and 2,000 test examples. The split seed is 20260520. Training uses date-coded seeds 20260521, 20260522, and 20260523, reported in the paper as logical seeds 21, 22, and 23. The primary Q1a v37 configuration uses AdamW with learning rate 2×10^{-5} , weight decay 0.01, a maximum of 10 epochs, bf16 precision, physical batch size 8 with four accumulation steps (effective batch size 32), gradient clipping 1.0, cosine scheduling, and a 0.20 warmup ratio. Early stopping patience is 10. The primary loss multipliers are 1.01 for implicit sentiment, 1.01 for sarcasm, 1.05 for irony, and 1.04 for code-switching; the other task multipliers are 1.0. The selection metric is development macro-pragmatic F1.

For each binary pragmatic head, the development threshold is selected from 0.05 through 0.95 in steps of 0.01 by binary macro-F1, with ties resolved toward 0.5. Q3 applies this rule independently for each seed and budget. Q4 uses raw positive-class sigmoid probabilities and therefore does not threshold the probabil-

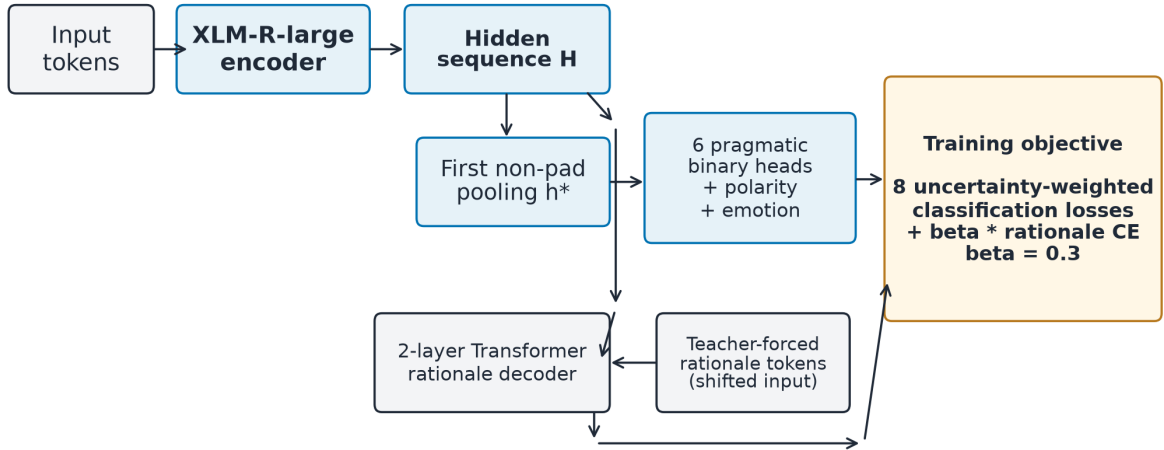


Figure 1: ViPragSent architecture and training objective. XLM-R-large produces a hidden sequence that feeds first-nonpadding pooling for six pragmatic binary heads, three-way polarity, and seven-way emotion prediction. The same hidden sequence is used as memory by a two-layer rationale decoder with teacher-forced shifted rationale tokens during training only; its rationale loss contributes to the objective and is not an inference output.

ities before computing ECE.

For Q2, polarity ECE is the ten-bin top-label ECE computed from the saved development predictions and development metrics for each seed, then reported on the 10^3 scale. The legacy field name `polarity_dev_ece` is retained for provenance; it does not indicate that the test-file value was used.

3.5 Provenance and comparison scope

The Q1a evidence package contains 18 saved prediction JSONL files: three ViPragSent target runs and three runs for each of five complete baseline families. Every one of the 45 target–baseline pairs has 2,000 rows with the same sample-ID set and the same six pragmatic gold-label tuples when aligned by sample ID. The Q1a scores were deterministically recomputed from those saved rows using the metric above. This supports a direct score comparison for the evaluated ID/gold cohort. It does not claim identical upstream input text, identical training, a significance test, or an independent rerun.

4 Experimental Setup

4.1 Separate experimental cohorts

Q1a, Q2, Q3, and Q4 are related but distinct artifact cohorts. The distinction is material when interpreting the full-model values.

The Q2 full value is not a rerun of Q1a v37: the cohorts differ in scheduler, warmup ratio, and early-stopping patience. Q2 is therefore interpreted within its own follow-up cohort, not as an explanation of the difference between Q1a’s 93.67 and Q2’s 92.8.

4.2 Data and external evaluation

Q1a evaluates the 2,000-example pragmatic test cohort. Q1b reports retention on UIT-VSFC, UIT-VSMEC, and AIVIVN. The external aggregate is called ordinary F1 in this paper: it is the unweighted mean of the three external macro-F1 values. It is not ordinal F1 and does not encode an ordinal relation among labels. External datasets have different domains and label semantics, so their scores are not pooled with the primary macro-pragmatic F1.

Counts are declared split counts where the inspected manifests provide them; * means that only the supplied official test-file count is evidenced. The original AIVIVN 2019 binary Kaggle files are provenance-only; Q1b evaluates the bundled derived three-way split. The machine-readable manifests retain the source ZIP SHA-256 23150E...32844AC2, ViPragSent package fingerprint 7C39...F4205EB, AIVIVN normalized-test hash 27DA...394EDA8,

Cohort	Run family and purpose	Key protocol
Q1a	Optimized v37 ViPragSent and five complete baseline families	Cosine scheduler, warmup 0.20, patience 10, development macro-pragmatic-F1 selection; target task multipliers as above
Q2	XLM-R follow-up full model and five ablations	Linear scheduler, warmup 0.10, patience 2; named auxiliary and uncertainty variants
Q3	XLM-R follow-up nested positive-label budgets	Linear scheduler, warmup 0.10, patience 2; development sarcasm binary macro-F1 selection
Q4	Calibration extraction	Same-seed extraction from each Q1a v37 full checkpoint; no new training

Experimental cohorts and protocols; quantitative entries elsewhere summarize three saved runs where reported ($n = 3$, mean $\pm$ sample SD).

UIT-VSFC hash C27B...2E75D96, and UIT-VSMEC hash 1D2E...3D955. Hashes are scope-specific evidence rather than license verification. The table reports dataset/package validation and saved artifact analysis, not an independent experiment rerun.

4.3 Baselines and reporting conventions

ViPragSent XLM-R-large is the only primary model. Comparison records include standard XLM-R, PhoBERT single-task and fine-tuned variants, Sailor, and Vistral where complete three-seed summaries are available. Vistral ViPragSent variants and incomplete GPT records are outside the primary Q1a comparison. ViSoBERT is contextual prior work, not an evaluated baseline in this study.

The available baseline manifests describe family-specific recipes rather than one recipe shared with the target. The standard XLM-R baseline uses AdamW at 2×10^{-5} , bf16, physical batch size 8 with four accumulation

steps, linear scheduling with 0.10 warmup, and a ten-epoch maximum; its selection metric is development macro-pragmatic F1. The PhoBERT fine-tune uses AdamW at 2×10^{-5} , bf16, physical batch size 32, linear scheduling with 0.10 warmup, and the same ten-epoch maximum. The PhoBERT single-task record is a bundle of six separately trained component artifacts and exposes no single aggregate optimizer recipe. Sailor-7B and Vistral-7B-Chat are causal-7B pragmatic-SFT records (Dou et al., 2024; Nguyen et al., 2023a) using paged AdamW 8-bit, learning rate 10^{-4} , bf16, physical batch size 2 with eight accumulation steps, cosine scheduling, 0.05 warmup, and a three-epoch maximum; their reported inference source is the classification heads. These records are described for provenance and interpretation only: the ViPragSent v37 cosine schedule, warmup, patience, and task multipliers are not claimed for the baselines.

Unless otherwise stated, a value is the arithmetic mean over three saved runs and the uncertainty is the sample standard deviation with $n = 3$. Q1a and Q3 report F1 in percentage points. Q2 reports macro-pragmatic F1, ECE multiplied by 10^3 , and relative cost. Q4 reports ECE on its native $[0,1]$ scale. Lower ECE is better. No significance test is applied to the three-seed summaries.

5 Results

5.1 Q1a: in-domain pragmatic performance

Table 1 gives the expanded canonical Q1a result with systems as rows and the six pragmatic metrics plus their macro average as columns. Values are mean $\pm$ sample SD in percentage points. The target remains highest in mean on every metric among the five complete three-seed baseline families; the other rows are contextual variants, provisional coverage, or one-shot records as marked in the caption and the machine-readable canonical row ledger.

Against the strongest complete baseline, the observed mean gaps are +0.51 pp for implicit sentiment, +1.24 pp for sarcasm, +0.50 pp for irony, +0.50 pp for idiom/figurative language, +0.57 pp for code-switching, and +1.07 pp for mocking. The two largest observed head gaps are therefore sarcasm and mocking, but these

Dataset / source	Task and labels	Declared split (train / dev / test)	Evaluation slice / split rule	Evidence and provenance status
ViPragSent_Experiment_Dataset_FINAL_V8	Vietnamese pragmatic sentiment; six binary pragmatic heads, separate 3-way polarity, and 7-way emotion	7,998 / 1,999 / 2,000	Q1a: 2,000-example test cohort; deterministic multilabel stratification; split seed 20260520	PACKAGE-VALIDATED; saved experiments ANALYZED; upstream SEACrowd/ViSoBERT artifact is labeled CC BY-NC 4.0; local derivative license not separately recorded
AIVIVN-human-derived-3way	Vietnamese sentiment; human-derived 3-way polarity (negative / neutral / positive)	12,869 / 1,609 / 1,609	Q1b: 1,609 held-out test records; project-defined split	ANALYZED; ID/gold/raw-text/NFC parity PASS; original Kaggle provenance reports Apache-2.0; manifest-hash discrepancy retained
UIT-VSFC	Vietnamese student feedback; 3-way polarity (0=negative, 1=neutral, 2=positive)	— / — / 3,166*	Supplied official test file; no project split seed	ANALYZED; observed ID/gold/text parity PASS; author-contact research access is private; no open license asserted
UIT-VSMEC	Vietnamese social-media emotion; 7 classes (anger, disgust, enjoyment, fear, other, sadness, surprise)	— / — / 693*	Supplied official test file; no project split seed	ANALYZED; observed ID/gold/text parity PASS; author-contact research access is private; no open license asserted

Current dataset statistics and evidence scope. Counts are declared train/dev/test split counts when available; an asterisk marks a supplied official test-file count, an em dash denotes an unavailable split count, and license status is reported only when present in the inspected evidence.

Model	Implicit sentiment	Sarcasm	Irony	Idiom/figurative	Code-switching	Mocking	Macro-pragmatic F1
ViPragSent (ours, XLM-R-large)	92.67 ± 0.24	89.36 ± 0.90	98.43 ± 0.20	93.35 ± 0.66	94.84 ± 0.56	93.35 ± 0.73	93.67 ± 0.19
PhoBERT (single-task)	88.00 ± 1.29	70.87 ± 1.34	97.13 ± 1.06	91.21 ± 0.78	91.08 ± 1.51	79.16 ± 2.99	86.24 ± 0.49
PhoBERT fine-tune	85.76 ± 0.72	72.64 ± 0.50	97.45 ± 0.19	86.63 ± 0.66	91.73 ± 0.53	78.53 ± 1.01	85.46 ± 0.36
XLM-R-large fine-tune	92.16 ± 0.13	88.12 ± 0.43	97.93 ± 0.10	92.85 ± 0.68	94.27 ± 0.04	92.27 ± 0.41	92.93 ± 0.14
Sailor-7B SFT	85.42 ± 0.27	72.14 ± 2.96	95.75 ± 0.52	91.15 ± 1.40	91.03 ± 0.74	77.76 ± 1.21	85.54 ± 0.48
Vistral-7B SFT	87.36 ± 0.31	74.74 ± 0.94	96.69 ± 0.23	91.48 ± 1.55	92.80 ± 0.60	79.91 ± 0.26	87.16 ± 0.33
ViPragSent - no auxiliary loss	87.43 ± 0.62	74.79 ± 0.53	97.15 ± 0.27	91.51 ± 0.99	92.88 ± 0.63	79.51 ± 2.81	87.21 ± 0.58
Vistral CoT-only clean rerun 003 (2/3)	17.36 ± 0.00	48.24 ± 0.00	48.20 ± 0.00	48.17 ± 0.00	45.92 ± 0.00	46.91 ± 0.00	42.47 ± 0.00
ViPragSent - explanation only	44.13 ± 0.00	48.24 ± 0.00	48.20 ± 0.00	48.17 ± 0.00	46.02 ± 0.18	46.91 ± 0.00	46.95 ± 0.03
ViPragSent (ours, Vistral)	84.93 ± 5.49	72.59 ± 7.73	94.75 ± 3.18	79.38 ± 22.13	78.36 ± 26.44	77.15 ± 8.60	81.19 ± 12.23
GPT-4.1-mini zero-shot	45.44 (n=1)	56.51 (n=1)	53.17 (n=1)	52.45 (n=1)	62.42 (n=1)	54.69 (n=1)	54.11 (n=1)
GPT-4.1-mini 8-shot	49.04 (n=1)	61.09 (n=1)	52.37 (n=1)	50.87 (n=1)	56.89 (n=1)	54.32 (n=1)	54.10 (n=1)

Table 1: Q1a in-domain pragmatic performance on the 2,000-example ID/gold test split. Systems are rows; the six pragmatic metrics and macro-pragmatic F1 are columns. The primary row is ViPragSent (ours, XLM-R-large). Entries are mean ± sample SD from the compact canonical rows; n=1 rows have no SD and the CoT row is provisional at n=2. Canonical row statuses are retained in the machine-readable q1a-extra ledger.

are descriptive differences across saved predictions, not significance claims. The macro difference is $93.665838 - 92.934251 = +0.731587$ percentage points, reported as +0.73 pp.

The expanded remote candidate coverage remains in Appendix B. The canonical row status fields are retained in the machine-readable ledger, and the two cohorts are kept separate because the report has a different cohort for some rows; its bootstrap confidence-interval half-widths are not used as sample SD.

5.2 Q1b: external retention

Table 2 expands Q1b using the canonical external-retention artifact. It shows mixed

transfer rather than uniformly preserved performance: the primary is not the highest ordinary-F1 record in this analyzed set, and its per-dataset profile differs from the classification baselines. The table retains the actual saved recipe names for the six comparable three-seed records and adds the complete three-seed PhoBERT-backed ViPragSent inventory as a contextual variant. The contextual row is not the primary model and is not a substitute for a reference-row recipe. The identity-mismatched Azure deployment and unresolved GPT-4o-mini 8-shot record remain in the machine-readable audit ledger and are not printed as manuscript rows.

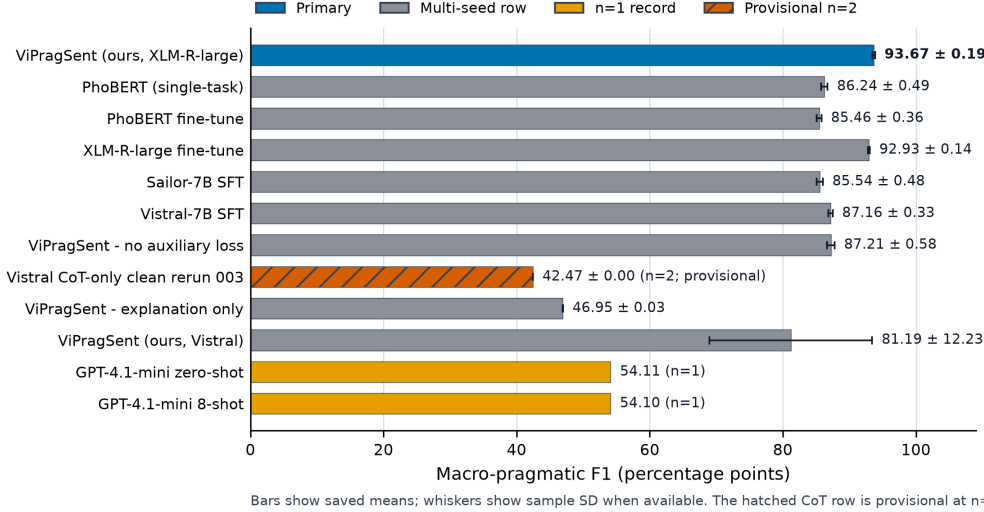


Figure 2: Q1a macro-pragmatic F1 across the expanded rows in Table 1. Bars show saved means on the 0–100 percentage-point scale; whiskers show sample SD when a multi-seed estimate is available. Complete three-seed rows are directly comparable for the primary baseline claim; one-shot reference records and the provisional $n = 2$ CoT record are retained for coverage and are not treated as complete three-seed baselines.

Model record	UIT-VSFC	UIT-VSMEC	AIVIVN-human-derived-3way	Ordinary F1
PhoBERT single-task routed	29.7 ± 0.5	38.8 ± 1.5	45.4 ± 2.6	38.0 ± 0.3
PhoBERT multitask 8-head	33.6 ± 3.0	36.4 ± 0.7	44.7 ± 1.3	38.2 ± 1.5
XLM-R-large multitask 8-head	39.1 ± 1.1	40.2 ± 0.2	48.3 ± 1.2	42.5 ± 0.5
ViPragSent (ours, XLM-R-large)	37.9 ± 2.8	40.0 ± 0.7	47.9 ± 2.2	42.0 ± 1.8
ViPragSent full PhoBERT inventory (contextual)	34.1 ± 0.7	36.0 ± 0.3	45.5 ± 0.8	38.6 ± 0.5
Sailor-7B SFT	34.4 ± 3.2	40.7 ± 0.7	48.7 ± 1.4	41.3 ± 1.2
Vistral-7B SFT	30.7 ± 6.4	39.4 ± 1.5	49.0 ± 0.8	39.7 ± 2.5

Table 2: External retention on the UIT-VSFC, UIT-VSMEC, and AIVIVN test cohorts. The primary row is ViPragSent (XLM-R-large); the PhoBERT-backed inventory row is contextual rather than primary. Entries are mean $\pm$ sample SD over three saved runs. Columns report macro-F1 in percentage points and their unweighted ordinary-F1 aggregate.

All entries in the main table are mean plus/minus sample SD over three saved runs, including the contextual inventory row. The external datasets differ in domain and target semantics, so the table is a retention profile rather than a single generalization score. The primary-versus-standard descriptive comparison has observed ID/gold/raw-text/ NFC parity on all three evaluated cohorts, while the AIVIVN source-manifest hash discrepancy remains a provenance limitation; no full-row ordinary-F1 superiority claim is made.

The canonical recipe ledger is retained separately from the metric table: `phobert_pol_single` + `phobert_emo_single` for PhoBERT single-task routed; `phobert_multitask_8head`; `xlmr_multitask_8head`;

`xlmr_followup_full` for the primary; `sailor_multitask_8head`; `vistral_multitask_8head`; and `azure_gpt41_mini` dedicated prompts for the actual GPT-4.1-mini deployment.

5.3 Q2: auxiliary-task and uncertainty ablations

Table 3 reports the separate XLM-R follow-up cohort. Within this cohort, the full configuration is not the highest reported macro-pragmatic F1: removing task-uncertainty weighting gives 92.9 versus 92.8 for the full configuration. The full configuration has lower reported ECE than the no-uncertainty variant, while removing the explanation auxiliary reduces relative cost substantially and changes F1 and ECE. Removing the multitask bundle

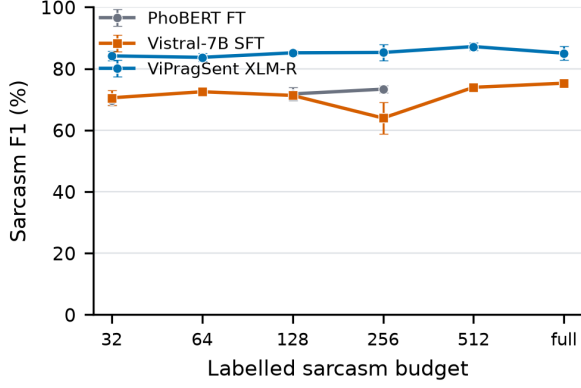


Figure 3: Q3 sarcasm F1 across pragmatic-label budgets. Curves show artifact-derived means and sample SD over three logical runs where the cohort is complete ($n = 3$; run labels 21, 22, and 23, corresponding to remote artifact seeds 20260521, 20260522, and 20260523). The PhoBERT curve has gaps at 64, 512, and full because those standard-GPU40 cells lack complete three-seed test evidence. The target is shown with PhoBERT fine-tune and Vistral-7B SFT for context.

causes a much larger F1 reduction. These are trade-offs within Q2, not evidence that one component is universally beneficial.

For compactness, Table 3 labels the variants as Full, – emotion, – explanation, Single-task bundle, – polarity, and – uncertainty; these mean the full follow-up, removal of the named auxiliary or weighting component, and the no-multitask component bundle, respectively.

Note: For all three no-polarity seeds, ECE is not applicable because the polarity head was removed; no polarity probability source exists. Table 3 uses the current Q2 protocol: ten-bin top-label ECE from the saved development metrics and predictions, reported on the 10^3 scale. The legacy field name `polarity_dev_ece` is retained for provenance. This is not Q4’s six-head calibration metric. The no-multitask entry is a separate single-task component-bundle recipe. Because Q2 is a separate cohort, its scientific comparison is the pattern across named variants, not an explanation of Q1a’s absolute score.

5.4 Q3: pragmatic label-budget behavior

Table 4 reports the XLM-R follow-up budget profile together with contextual PhoBERT and Vistral baseline curves. Complete entries are mean $\pm$ sample SD over three runs.

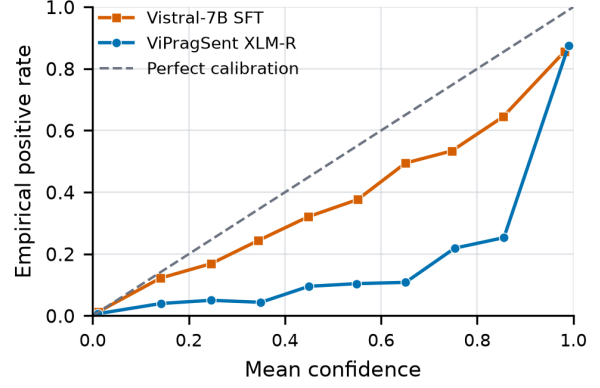


Figure 4: Q4 reliability diagrams for the calibration comparison. Curves aggregate the supplied per-label reliability bins over three logical runs ($n = 3$; run labels 21, 22, and 23, corresponding to remote artifact seeds 20260521, 20260522, and 20260523). The dashed line denotes perfect calibration; the figure is descriptive and does not imply target dominance.

Thresholds are tuned on the fixed development split; the displayed metrics are evaluated on the fixed test split. The standard GPU40 PhoBERT cohort has complete three-seed evidence only at 32, 128, and 256 positives; its 64, 512, and full-budget cells are not imputed. A separate compact-final PhoBERT cohort is documented in `figures/source_data/q3_phobert_cohort_audit.md` but is not mixed into this table because its 32 and 128 aggregates differ.

ViPragSent (XLM-R-large) exceeds the complete PhoBERT baseline rows at the budgets where that cohort is available. Its curves are non-monotonic: macro-pragmatic F1 rises from 91.1 at 32 positives to 92.3 at 512 and returns to 91.1 at full budget. Missing 64/512/full PhoBERT cells limit the contextual baseline curve. This is a sample-efficiency diagnostic, not evidence that less data is generally preferable; the supplied plot visualizes the same cohort.

5.5 Q4: calibration

Under the Q4 extraction protocol, the ViPragSent XLM-R target has $\text{ECE } 0.0386 \pm 0.0069$, while the Vistral comparison record has $\text{ECE } 0.0345 \pm 0.0041$. Lower ECE is better, so the comparison does not support target calibration dominance. Q4 extracts raw positive sigmoid probabilities from same-seed Q1a v37 checkpoints without retraining; ECE de-

XLM-R variant	Macro-pragmatic F1	Polarity dev	ECE ($\times 10^3$)	Relative cost
Full	92.8 $\pm$ 0.6		81.1 $\pm$ 11.7	1.00
- emotion	91.7 $\pm$ 0.7		56.2 $\pm$ 22.5	0.65
- explanation	92.3 $\pm$ 1.3		68.1 $\pm$ 22.2	0.20
Single-task bundle	74.7 $\pm$ 2.3		118.9 $\pm$ 69.1	1.10
- polarity	92.8 $\pm$ 0.2		not applicable	0.81
- uncertainty	92.9 $\pm$ 0.3		90.2 $\pm$ 5.9	0.82

Table 3: Q2 follow-up ablations. Macro-pragmatic F1 is evaluated on the Q2 test split; polarity ECE is the verified development-split ten-bin top-label ECE, shown on the $\times 10^3$ scale. The legacy field name is `polarity_dev_ece`; the no-polarity row is not applicable because its polarity head was removed. Entries are mean $\pm$ sample SD over $n = 3$ runs.

Positive examples	ViPragSent (XLM-R-large) sarcasm / macro	PhoBERT (fine-tune) sarcasm / macro	Vistral-7B SFT sarcasm / macro
32	84.2 $\pm$ 1.6 / 91.1 $\pm$ 1.3	69.7 $\pm$ 1.5 / 84.2 $\pm$ 0.7	70.5 $\pm$ 2.4 / 86.2 $\pm$ 0.2
64	83.7 $\pm$ 1.3 / 91.2 $\pm$ 0.8	–	72.5 $\pm$ 0.9 / 86.7 $\pm$ 0.5
128	85.2 $\pm$ 0.7 / 91.6 $\pm$ 0.2	71.8 $\pm$ 2.2 / 84.0 $\pm$ 1.2	71.3 $\pm$ 0.6 / 86.2 $\pm$ 0.4
256	85.3 $\pm$ 2.6 / 91.6 $\pm$ 1.2	73.4 $\pm$ 1.2 / 84.5 $\pm$ 0.9	64.0 $\pm$ 5.1 / 70.3 $\pm$ 14.2
512	87.2 $\pm$ 1.2 / 92.3 $\pm$ 0.7	–	73.9 $\pm$ 1.2 / 87.2 $\pm$ 0.3
Full	85.0 $\pm$ 2.3 / 91.1 $\pm$ 1.3	–	75.3 $\pm$ 0.8 / 87.4 $\pm$ 0.2

Table 4: Q3 pragmatic label-budget behavior on the fixed test split, using the three available systems in `figures/source_data/table_q3_low_resource.csv`: ViPragSent (XLM-R-large), PhoBERT (fine-tune), and Vistral-7B SFT. Complete entries are mean $\pm$ sample SD over $n = 3$ logical runs; – denotes PhoBERT cells without a complete three-seed test cohort and is not imputed. The separate compact-final PhoBERT cohort is not mixed with the standard GPU40 cohort.

pendes on its split, ten bins, and probability source.

6 Analysis and Discussion

6.1 What the primary result establishes

Q1a establishes a direct score comparison for the saved matched cohort: all 45 pairs share 2,000 sample IDs and six pragmatic gold tuples, and macro-F1 was deterministically re-computed. ViPragSent records 93.67 versus 92.93 for standard XLM-R (+0.73 pp), leads on all six heads, and has the largest observed gaps on sarcasm (+1.24 pp) and mocking (+1.07 pp). It has the highest recorded mean among the five complete baseline families, but the result remains descriptive because recipes differ.

6.2 Auxiliary supervision is an operating-point choice

Q2 shows that auxiliary supervision is an operating-point choice. The full variant has 92.8 macro-pragmatic F1, while removing uncertainty weighting is slightly higher; the other variants trade F1, ECE, and cost in different directions, with the no-multitask bundle showing a large F1 reduction. These three-run com-

parisons do not isolate a causal mechanism. Q2 is a separate cohort, not an explanation of Q1a’s 93.67.

The rationale component is auxiliary teacher-forced token supervision during training; classification heads remain the deployed interface. The results therefore support a representation-learning interpretation, not faithful natural-language explanations at inference.

6.3 Transfer and calibration expose different weaknesses

Q1b and Q4 keep in-domain performance from carrying the argument: external retention is mixed, and Vistral has lower ECE under the supplied Q4 protocol. Discrimination, transfer, and calibration answer different questions, so deployment selection should match the operating point rather than a single macro-F1 value.

6.4 Label budgets and variance

Q3 shows why positive-label budgets need run variation. The best displayed target mean occurs at 512 positives, while the full budget does not dominate; missing PhoBERT cells limit the contextual comparison. Threshold selection, class balance, and optimization vari-

ance may contribute, but the artifacts do not isolate them. Budget studies therefore need a fixed negative pool, explicit thresholds, contextual baselines, and uncertainty summaries.

7 Limitations

The paper analyzes saved artifacts rather than an independent training or inference rerun; deterministic recomputation verifies the Q1a arithmetic but not fresh-run reproducibility. Q1a is scoped to the matched ID/gold prediction cohort, while Q2 and Q3 are separate follow-up run families and Q4 extracts calibration from Q1a v37 checkpoints. External retention is finite and heterogeneous across UIT-VSFC, UIT-VSMEC, and AIVIVN, and ECE depends on its specified split, binning, and probability source. The operational pragmatic labels do not exhaust discourse or speaker intent, and the evidence does not establish group-conditional fairness, annotation history, consent, licensing, or institutional-review details.

8 Ethics and Responsible Use

The data handoff identifies a frozen ViPragSent package sourced through SEACrowd/ViSoBERT and records raw-text preservation, split sizes, and label fields. The current project record does not provide sufficient evidence for this paper to assert consent procedures, institutional review or exemption, compensation, licensing terms, or de-identification beyond those manifest fields. We make no such claims.

Vietnamese text with pragmatic and affective signals may contain offensive, personal, or socially sensitive language. Classification errors may mischaracterize speakers or amplify stereotypes. The model is therefore an analysis aid requiring human review, not an autonomous adjudicator of a person’s intent, emotion, credibility, or identity. Aggregate macro-F1 does not establish equal performance across social groups, domains, or writing styles. The manuscript does not reproduce individual example text, and the system is not intended for high-impact decisions or inference of protected attributes.

9 Conclusion

ViPragSent is an XLM-R-large multi-task model with separate pragmatic, polarity, and emotion heads and a training-only rationale objective. On the matched Q1a ID/gold cohort it records 93.67 ± 0.19 macro-pragmatic F1 versus 92.93 ± 0.14 for standard XLM-R (+0.73 pp), with the highest saved mean on all six heads among five complete baseline families. Q1b is mixed, Q2 exposes accuracy/calibration/cost trade-offs, Q3 is non-monotonic with contextual baselines, and Q4 shows no blanket calibration advantage. The primary result is descriptive and cohort-specific, but it supports the implemented model as a focused account of Vietnamese pragmatic prediction.

References

- Rich Caruana. 1997. [Multitask learning](#). *Machine Learning*, 28(1):41–75.
- Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishra Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. 2020. [Unsupervised cross-lingual representation learning at scale](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 8440–8451, Online. Association for Computational Linguistics.
- Longxu Dou, Qian Liu, Guangtao Zeng, Jia Guo, Jiahui Zhou, Xin Mao, Ziqi Jin, Wei Lu, and Min Lin. 2024. [Sailor: Open language models for south-East Asia](#). In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 424–435, Miami, Florida, USA. Association for Computational Linguistics.
- Debanjan Ghosh, Alexander R. Fabbri, and Smaranda Muresan. 2018. [Sarcasm analysis using conversation context](#). *Computational Linguistics*, 44(4):755–792.
- Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. 2017. [On calibration of modern neural networks](#). *Proceedings of Machine Learning Research*, 70:1321–1330.
- Vong Anh Ho, Duong Huynh-Cong Nguyen, Danh Hoang Nguyen, Linh Thi-Van Pham, Duc-Vu Nguyen, Kiet Van Nguyen, and Ngan Luu-Thuy Nguyen. 2020. [Emotion recognition for Vietnamese social media text](#). In *Computational Linguistics: 14th International Conference of*

the Pacific Association for Computational Linguistics, *PACLING 2019, Revised Selected Papers*, volume 1215 of *Communications in Computer and Information Science*, pages 319–333. Springer.

Cheng-Yu Hsieh, Chun-Liang Li, Chih-kuan Yeh, Hootan Nakhost, Yasuhisa Fujii, Alex Ratner, Ranjay Krishna, Chen-Yu Lee, and Tomas Pfister. 2023. [Distilling step-by-step! outperforming larger language models with less training data and smaller model sizes](#). In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 8003–8017, Toronto, Canada. Association for Computational Linguistics.

Alex Kendall, Yarin Gal, and Roberto Cipolla. 2018. [Multi-task learning using uncertainty to weigh losses for scene geometry and semantics](#). In *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 7482–7491. IEEE.

Tao Lei, Regina Barzilay, and Tommi Jaakkola. 2016. [Rationalizing neural predictions](#). In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, pages 107–117, Austin, Texas. Association for Computational Linguistics.

Chien Van Nguyen, Thuat Nguyen, Quan Nguyen, Huy Nguyen, Björn Plüster, Nam Pham, Huu Nguyen, Patrick Schramowski, and Thien Nguyen. 2023a. [Vistral-7B-Chat – towards a state-of-the-art large language model for Vietnamese](#). Hugging Face model card. Primary model-release documentation and citation supplied by the authors.

Dat Quoc Nguyen and Anh Tuan Nguyen. 2020. [PhoBERT: Pre-trained language models for Vietnamese](#). In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 1037–1042, Online. Association for Computational Linguistics.

Kiet Van Nguyen, Vu Duc Nguyen, Phu X. V. Nguyen, Tham T. H. Truong, and Ngan Luu-Thuy Nguyen. 2018. [UIT-VSFC: Vietnamese students’ feedback corpus for sentiment analysis](#). In *2018 10th International Conference on Knowledge and Systems Engineering (KSE)*, pages 19–24. IEEE.

Nam Nguyen, Thang Phan, Duc-Vu Nguyen, and Kiet Nguyen. 2023b. [ViSoBERT: A pre-trained language model for Vietnamese social media text processing](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 5191–5207, Singapore. Association for Computational Linguistics.

A Reproducibility record

The following record is part of the manuscript rather than a future action list. All paths are

relative to the paper workspace unless a source path is explicitly identified.

Primary backbone FacebookAI/xlm-roberta-large, revision c23d21b0620b635a76227c604d44e43a9f0ee389

Model interface First nonpadding-token pooling; six binary pragmatic heads, three-way polarity head, seven-way emotion head; classification heads are the inference source

Rationale implementation Two-layer causal TransformerDecoder with memory projection; hidden 128, four heads, feed-forward 512, dropout 0.1; tied token embeddings; teacher forcing; beta 0.3; target maximum length 160; decoder discarded at inference

Rationale target provenance Target manifest artifact approved_generated_rationales_train.jsonl; 7,998 training rows; provider unspecified in the inspected evidence; not treated as original annotation-history evidence

Dataset package SEACrowd/ViSoBERT source package; 11,997 rows; train/dev/test = 7,998/1,999/2,000; split seed 20260520; local processed fingerprint 7C39BEEBC462D1F9076F5DF1565E924BD884263D5503F23398BD83A6BF4205EB

Q1a target artifact provenance Remote target dataset fingerprint B906C090400BAE115C9C5E3C35E32FA410AC519AE09209EBA741F198087C24F9

Q1a baseline artifact provenance Remote baseline dataset fingerprint A13573E38550ABD55D7F63E983C602BEA13C2765A1FFECA285F718478532AF0D

Training seeds Date-coded seeds 20260521, 20260522, 20260523; logical labels 21, 22, 23

Primary Q1a optimizer AdamW, learning rate 2e-5, weight decay 0.01, bf16, physical batch 8, accumulation 4, effective batch 32, maximum 10 epochs, patience 10, gradient clipping 1.0

Primary Q1a schedule Cosine, warmup ratio 0.20, development macro-pragmatic-F1 selection; task multipliers implicit 1.01, sarcasm 1.01, irony 1.05, code-switching 1.04, remaining tasks 1.0

Follow-up Q2/Q3 schedule AdamW, learning rate 2e-5, bf16, physical batch 8, accumulation 4, effective batch 32, maximum 10 epochs, linear schedule, warmup ratio 0.10, patience 2

Threshold rule Candidate thresholds 0.05–0.95, step 0.01; development binary macro-F1; ties toward 0.5

Q3 mask rule Nested positive subsets of 32, 64, 128, 256, 512, and full; fixed negative pool; positive weights recomputed per budget; development threshold per seed/budget

Q4 rule Six pragmatic heads; raw positive sigmoid probabilities; ten equal-width bins; no temperature scaling; test split; mean and sample SD across seeds; no new training

Q1a evidence evidence/q1a_fairness_comparison.json, evidence/q1a_fairness_run_table.csv, evidence/q1a_verified_table_inputs.json, evidence/q1a_numeric_claim_ledger.json

Method evidence evidence/method_evidence.json, evidence/evidence_claim_ledger.json

Protocol lineage evidence/protocol_lineage.json; records distinct Q1a v37, Q2, Q3, and Q4 run families and Q4 source-checkpoint lineage

Prediction rows raw_fairness/*.jsonl; 18 files, 2,000 rows per file; deterministic metric recomputation matches recorded values

The Q1a comparison covers 45 target–baseline pairs across the five complete baseline families. Each pair has the same sample-ID set and six-label gold tuples after alignment by ID. The comparison is descriptive over saved predictions. No independent training or inference rerun is part of this record. The three

fingerprints above are retained as separate local and remote provenance identifiers; the direct score comparison is scoped to the verified same-ID, same-gold cohort rather than treated as a fingerprint-equality test.

B Table-specific protocol and missingness

Q1a uses the optimized v37 target and the complete three-seed baseline families listed in Table 1. The target and standard XLM-R macro means are 93.665838 and 92.934251, with sample SDs 0.187121 and 0.144790 percentage points. Their difference is 0.731587 percentage points. The per-head means and SDs in Table 1 were recomputed from the same saved JSONL prediction rows.

Q1b uses the external retention records and reports ordinary F1 as the unweighted mean of the three external macro-F1 values. Q2 uses the six follow-up variants: full, no emotion auxiliary, no polarity auxiliary, no rationale, no multitask, and no uncertainty weighting. The selected Q2 metric artifacts expose the legacy field `polarity_dev_ece`; Table 3 reports macro-pragmatic F1 on the Q2 test split and the verified ten-bin top-label polarity ECE from the development split on the 10^3 scale. The no-polarity row is not applicable for all three seeds because its polarity head was removed. The no-multitask row is a separate single-task component-bundle recipe. Relative cost is normalized to the full XLM-R Q2 run using the recorded mean GPU-hour values. Q3 reports the six nested target budgets and the available contextual baseline rows; missing baseline budget cells are not imputed. Q4 extracts raw probabilities from the same-seed Q1a v37 full checkpoints.

Audit note: the corresponding test-file values are valid test ECE, but the prior paper version incorrectly labeled them as development ECE. This revision corrects that split label and uses the verified development values in Table 3.

For auditability, the no-multitask component-bundle manifest records `status = NOT_STARTED` with `execution_kind = component_bundle`, direct classification outputs used, synthetic results false, and review status PASS. These technical flags identify

the recipe protocol and are not a failed-run claim; the component execution proof is the basis for retaining its reported metrics.

Incomplete GPT records and ViSoBERT prediction records are not inserted into any table. The retained ViPragSent Vistral variants are shown only where the compact canonical rows provide an explicit evidence status; the incomplete CoT-only row remains marked provisional. Q2 ECE for the no-polarity row is reported as not applicable for all three seeds because the polarity head was removed. These exclusions define the comparison scope and do not support claims about systems without verified rows.

The exact Hugging Face seed-22 run directory is present in the current `overflow-006` and Vistral-checkpoint trees. It contains development predictions/metrics and checkpoint/receipt artifacts, but the exhaustive live tree audit found no test predictions or test metrics for this exact run. The development metric is not substituted for a test score, so the COT row remains provisional at 2/3 test seeds. The detailed audit is recorded in `revision_q1a_extra/q1a_cot_seed22_hf_audit.json`.